

LEC climatology figures: before versus after

A direct comparison of the published legacy calculation and the validated LorenzCycleToolKit 2.0.0 rerun for Southwestern Atlantic cyclones

Paired cyclones	3,820
Primary rows per version	15,280
Toolkit commit	d38cda7e37d8
Correction commit	d07707767c29
Comparison files	20 PNG + 20 PDF

Purpose

This document places the legacy before result and the corrected after result in the same figure wherever possible. Both sides use the identical 3,820-cyclone population, so the differences isolate the toolkit correction rather than a change of sample. Published article files remain untouched.

Scope decisions

The main panels use one primary incipient, intensification, mature and decay record per cyclone. Secondary lifecycle episodes remain preserved upstream but are outside this four-phase comparison. EOFs use the published 24-term correlation-matrix definition; corrected modes are matched to legacy modes by loading-pattern correlation and sign-aligned. Intense groups retain the published 90th-percentile criterion and are matched by standardized centroid distance.

Before-after numerical overview

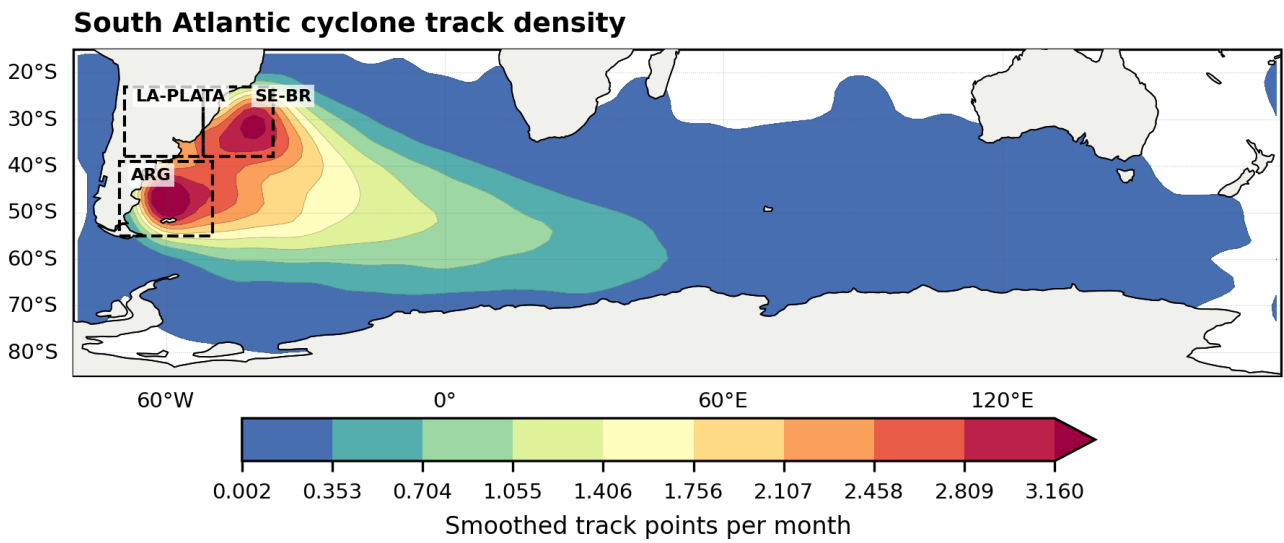
Phase	Ca before -> after	Ck before -> after	Ce before -> after	EOF1 variance
incipient	0.69 -> 1.92	-1.82 -> 0.33	3.80 -> 3.80	21.0% -> 20.6%
intensification	1.40 -> 4.12	-4.15 -> -0.30	6.23 -> 6.23	30.1% -> 27.5%
mature	1.18 -> 3.67	-5.71 -> -2.43	4.55 -> 4.55	28.6% -> 26.3%
decay	0.68 -> 2.16	-4.00 -> -2.10	1.98 -> 1.98	31.2% -> 29.2%

Intense-cyclone groups

The pointwise vorticity threshold is 8.293; 1,486 paired cyclones enter both four-group analyses. After groups are reordered to their closest before centroid.

Version	Cluster	n	Mean max. vorticity	Median max. vorticity
before	1	745	9.82	9.60
before	2	421	10.84	10.61
before	3	193	10.14	9.86
before	4	127	11.04	11.01
after	1	745	9.82	9.60
after	2	421	10.84	10.61
after	3	193	10.14	9.86
after	4	127	11.04	11.01

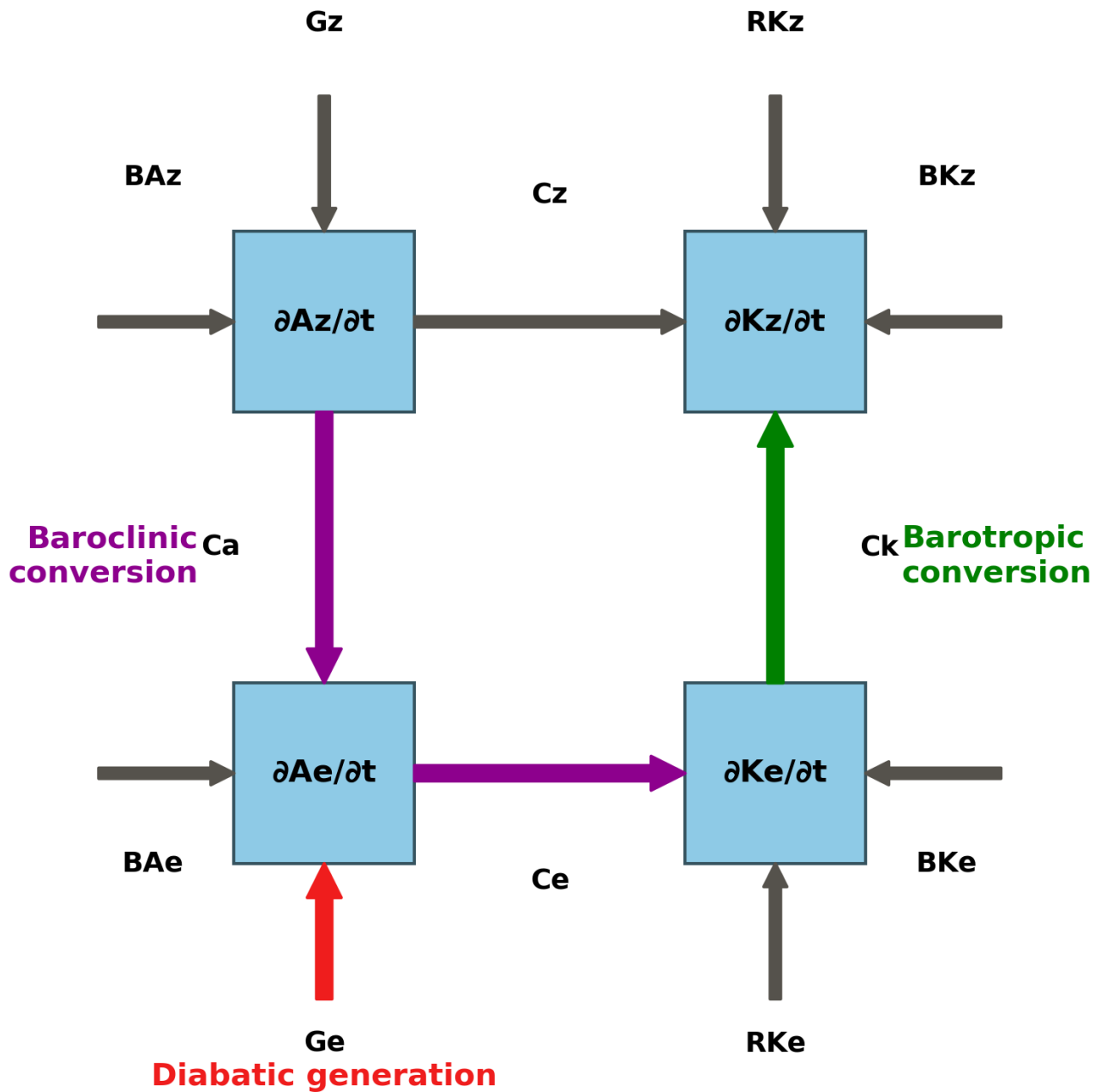
Figure 1



Track density and the three genesis regions; this track-only reference is common to both versions.

Figure 2

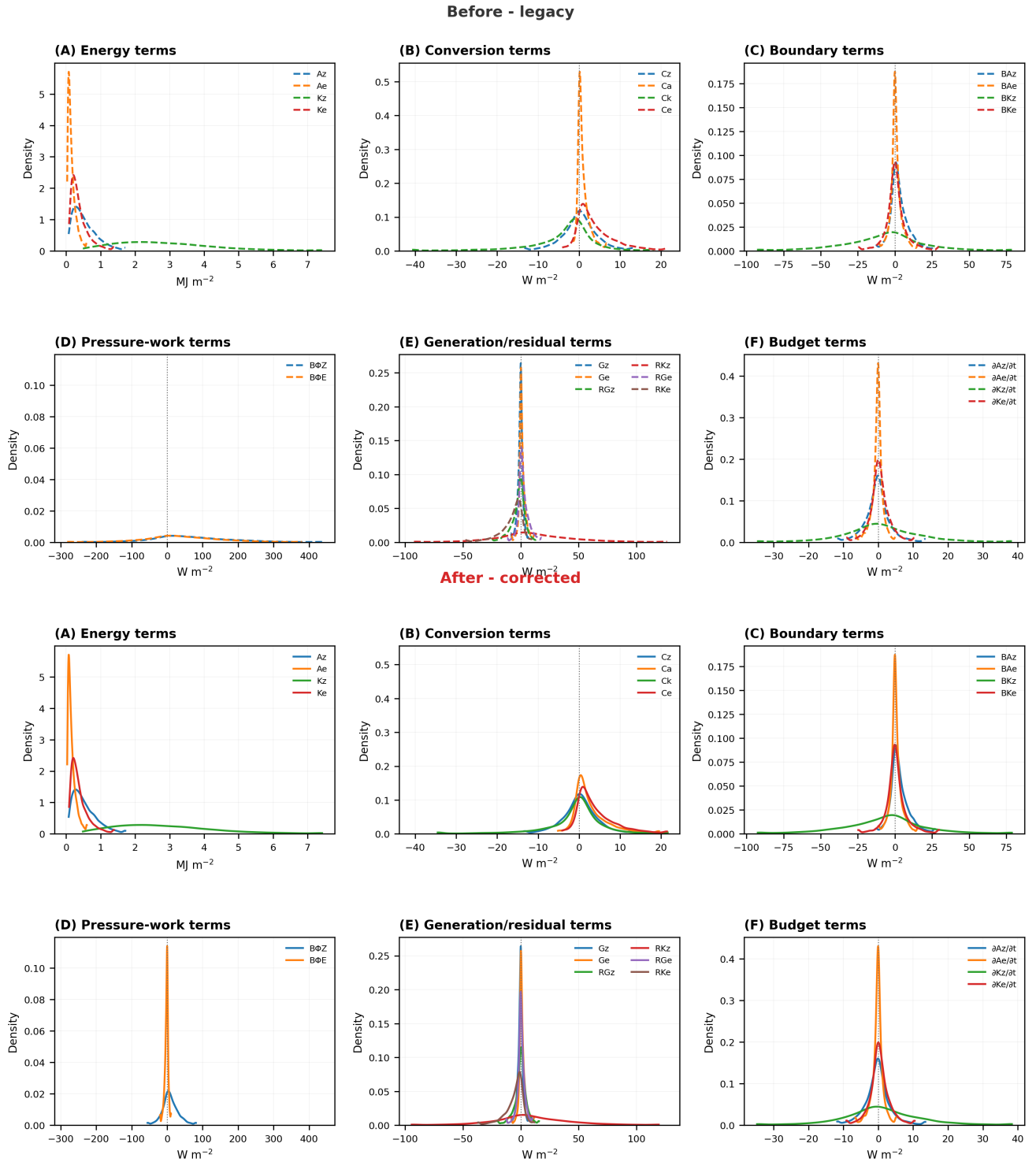
Lorenz Energy Cycle reference diagram



Reference four-box Lorenz Energy Cycle diagram; this conceptual panel is common to both versions.

Figure 3

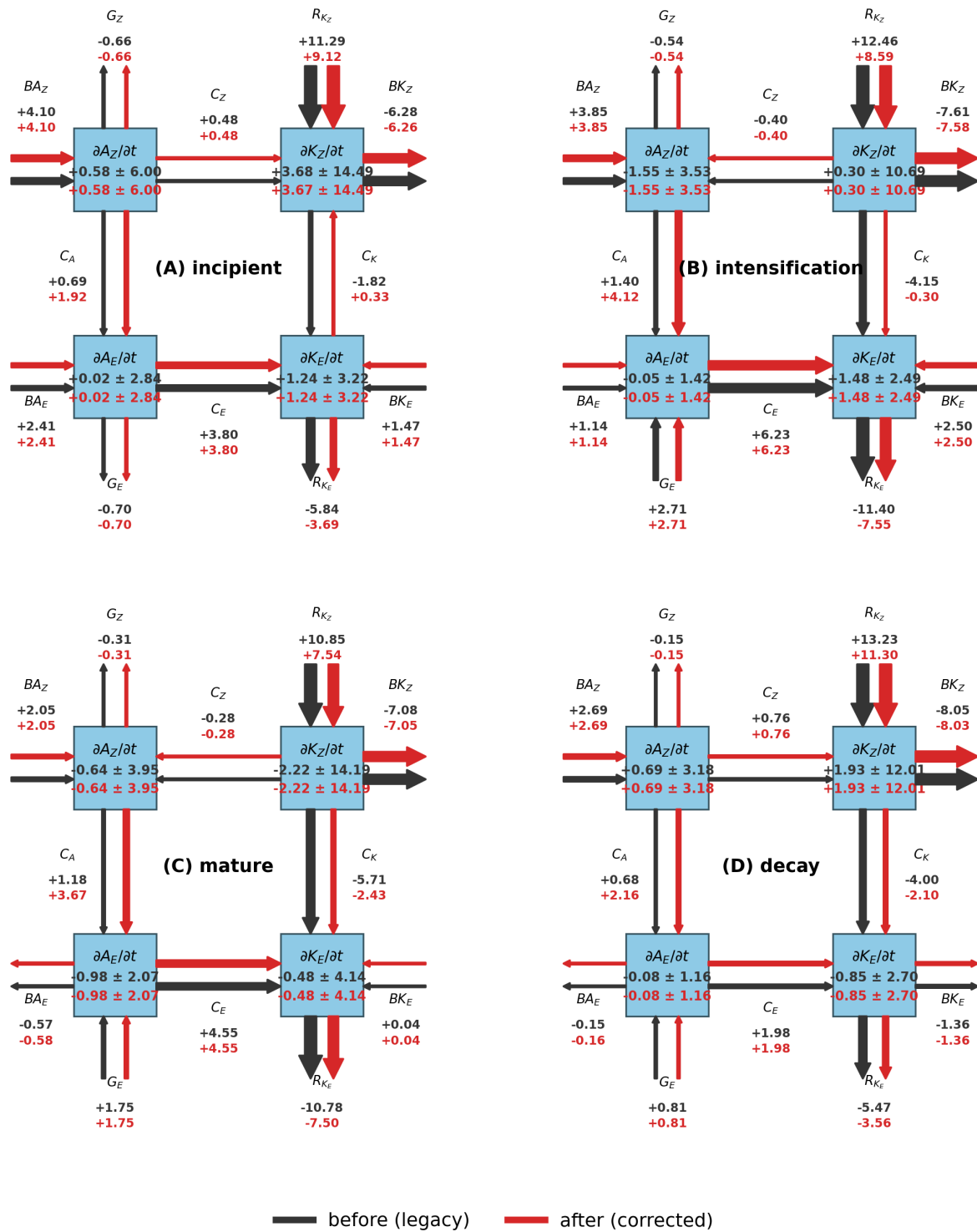
LEC term distributions before and after the toolkit correction



LEC term probability densities with the legacy before distributions above and corrected after distributions below.

Figure 4

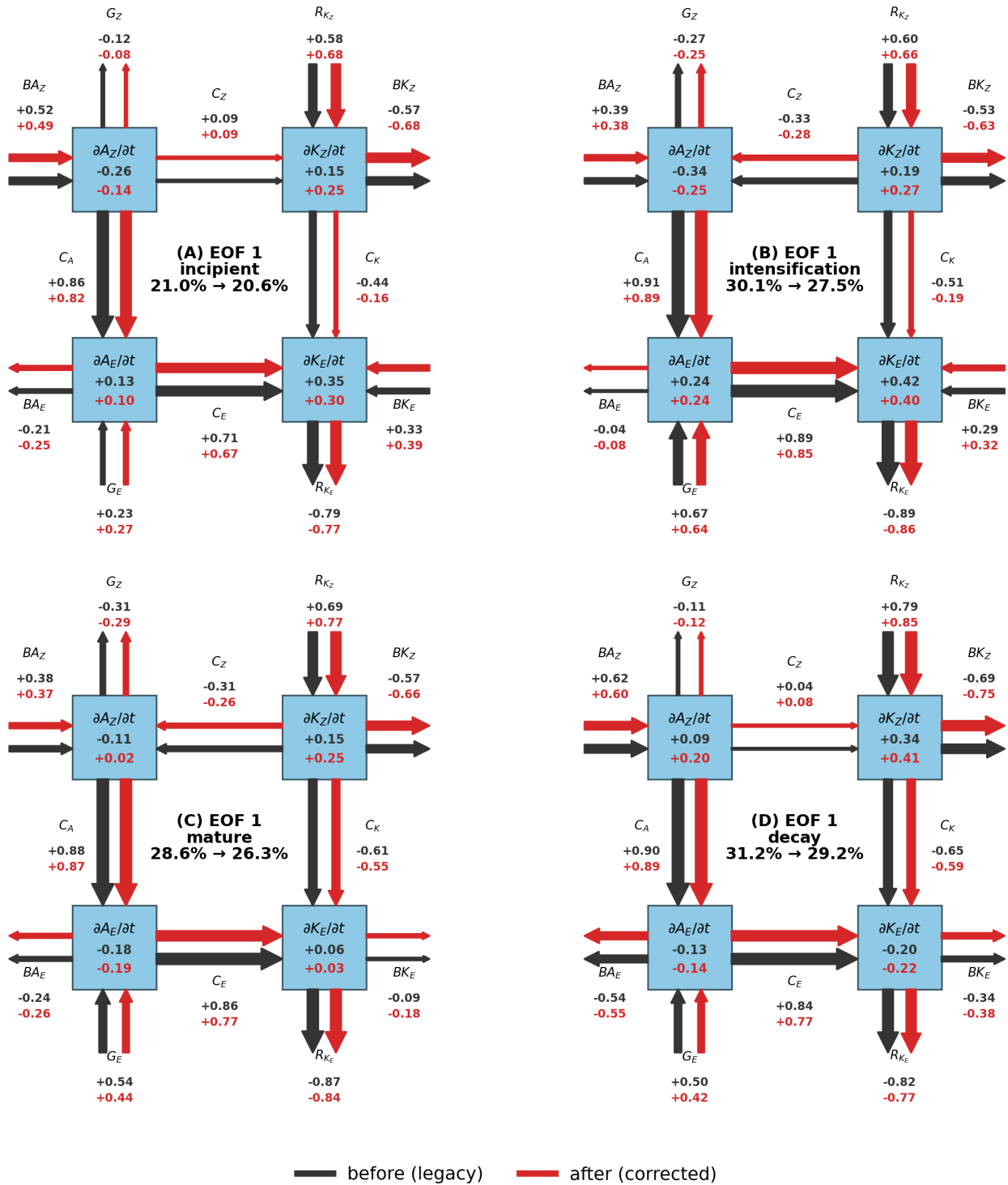
Phase-mean Lorenz Energy Cycle before and after mean $\pm$ sample standard deviation



Phase-mean LEC: dark arrows and values are before; red arrows and values are after.

Figure 5

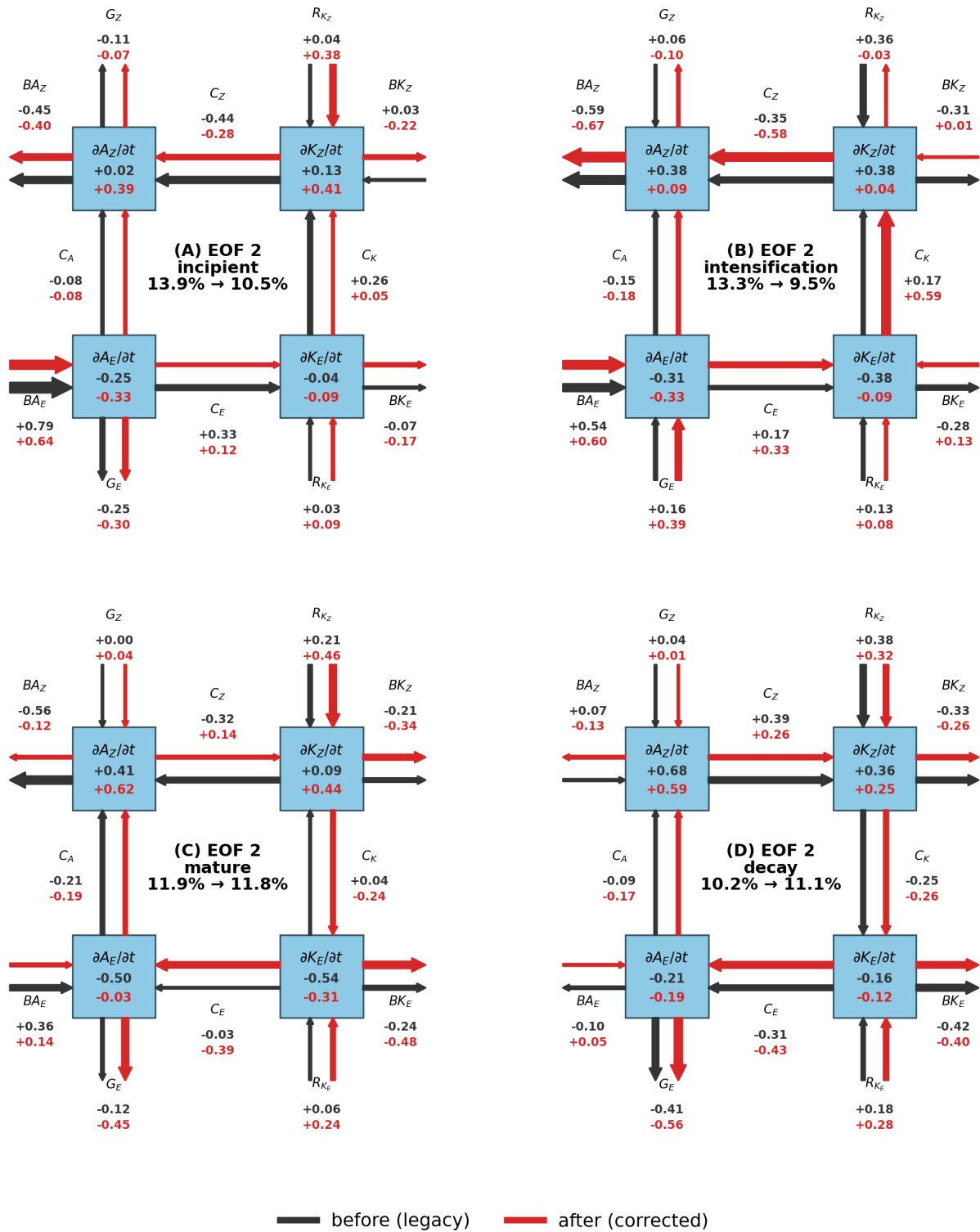
EOF 1 loadings before and after



EOF 1 LEC loadings before and after, after matching modes and aligning their signs.

Figure 6

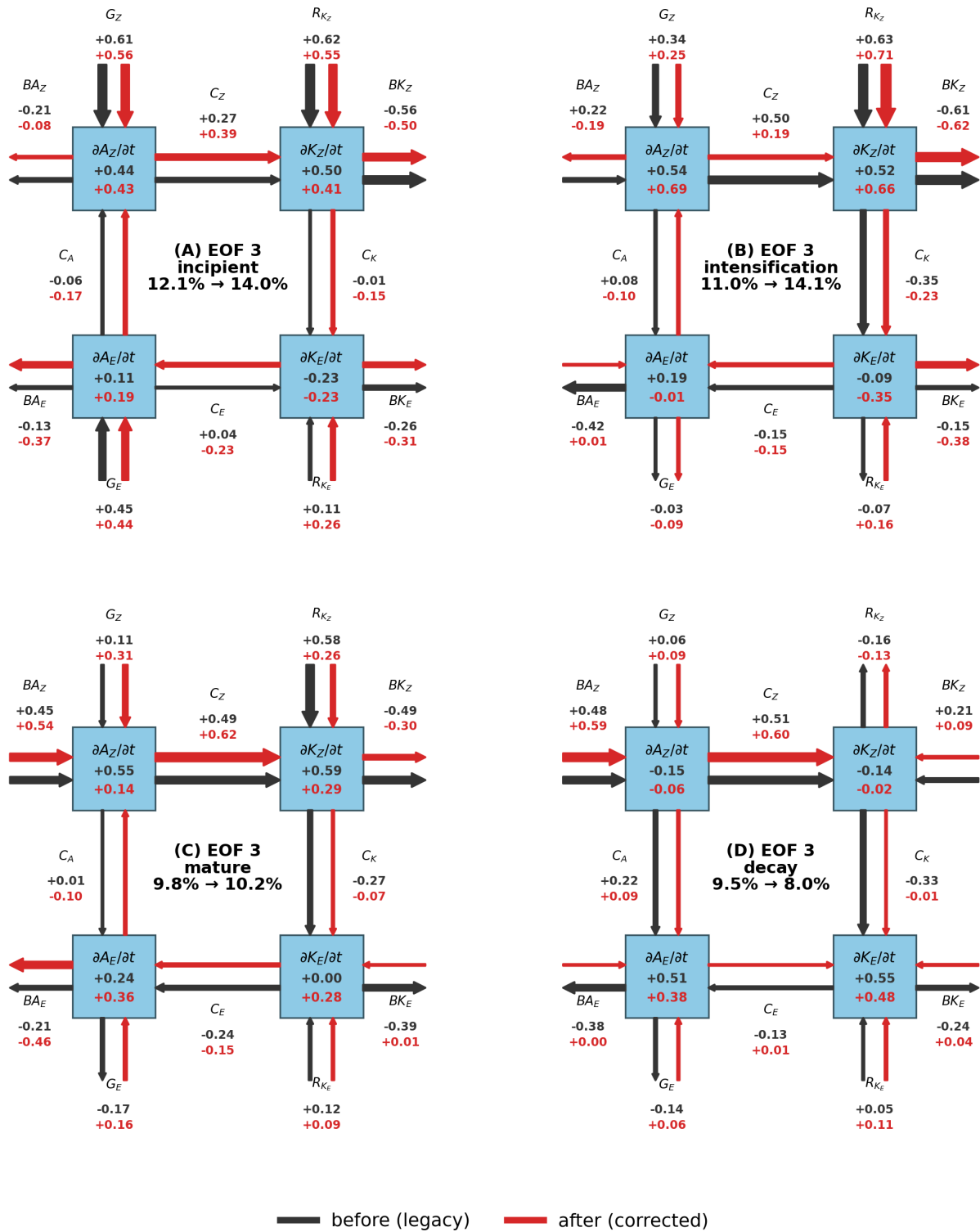
EOF 2 loadings before and after



EOF 2 LEC loadings before and after, after matching modes and aligning their signs.

Figure 7

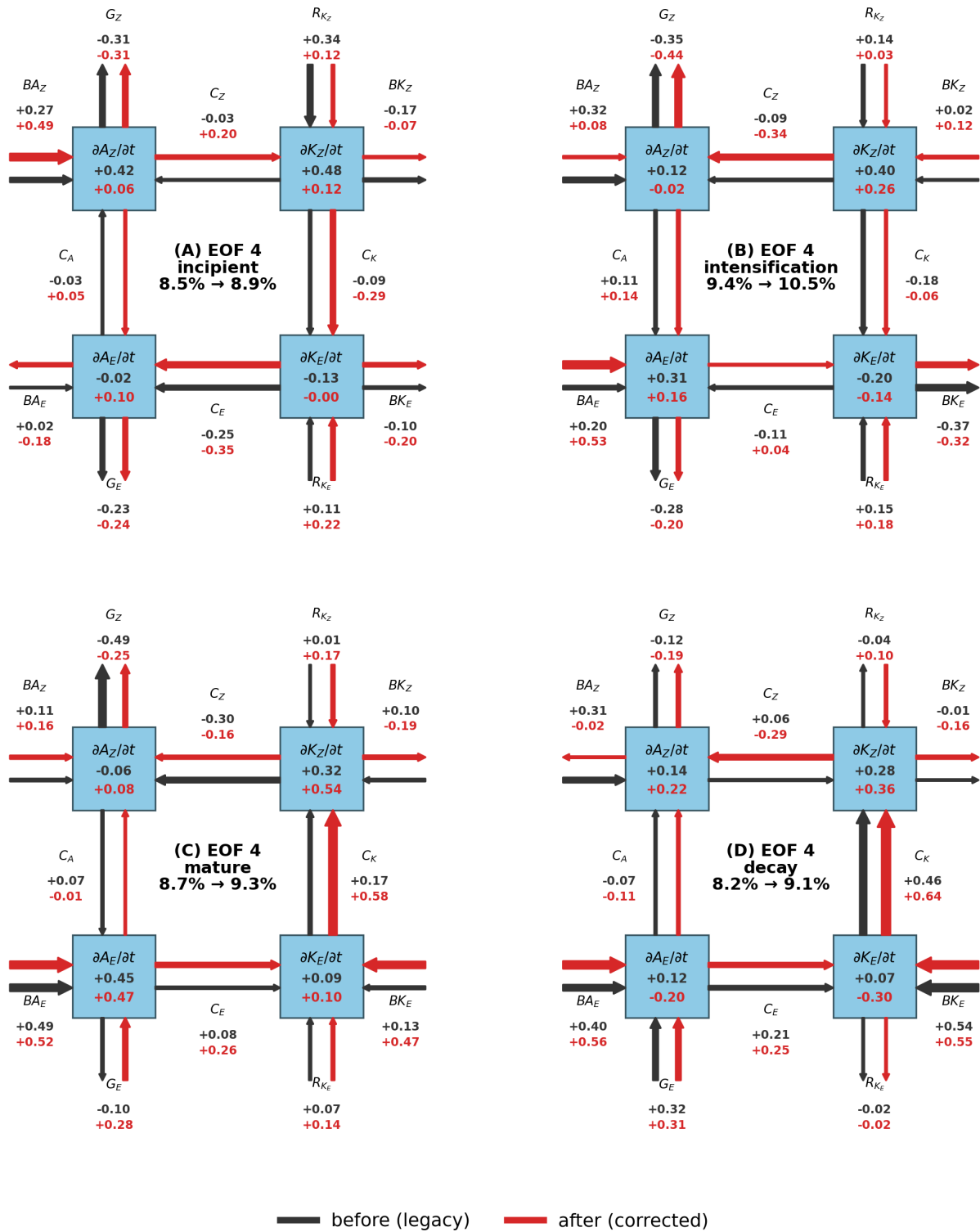
EOF 3 loadings before and after



EOF 3 LEC loadings before and after, after matching modes and aligning their signs.

Figure 8

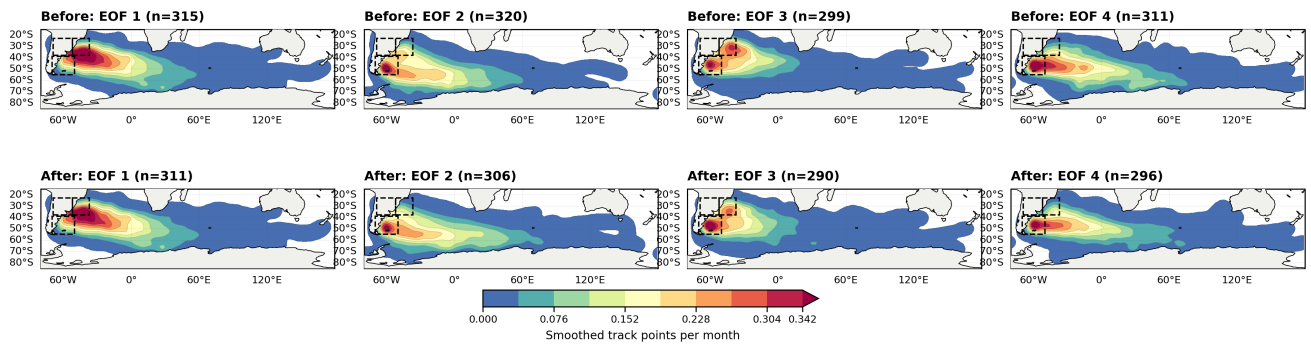
EOF 4 loadings before and after



EOF 4 LEC loadings before and after, after matching modes and aligning their signs.

Figure 9

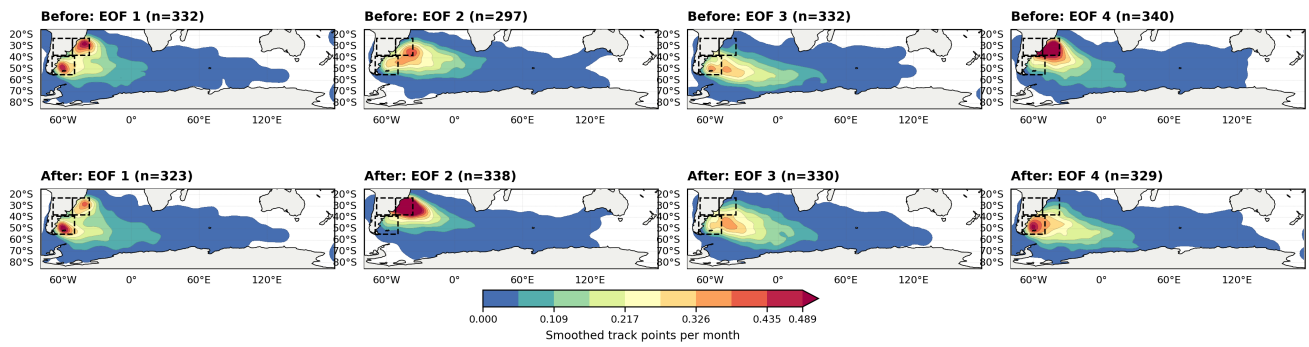
Track density - positive total-lifecycle PC extremes



Positive PC-extreme track densities with before on the upper row and after on the lower row.

Figure 10

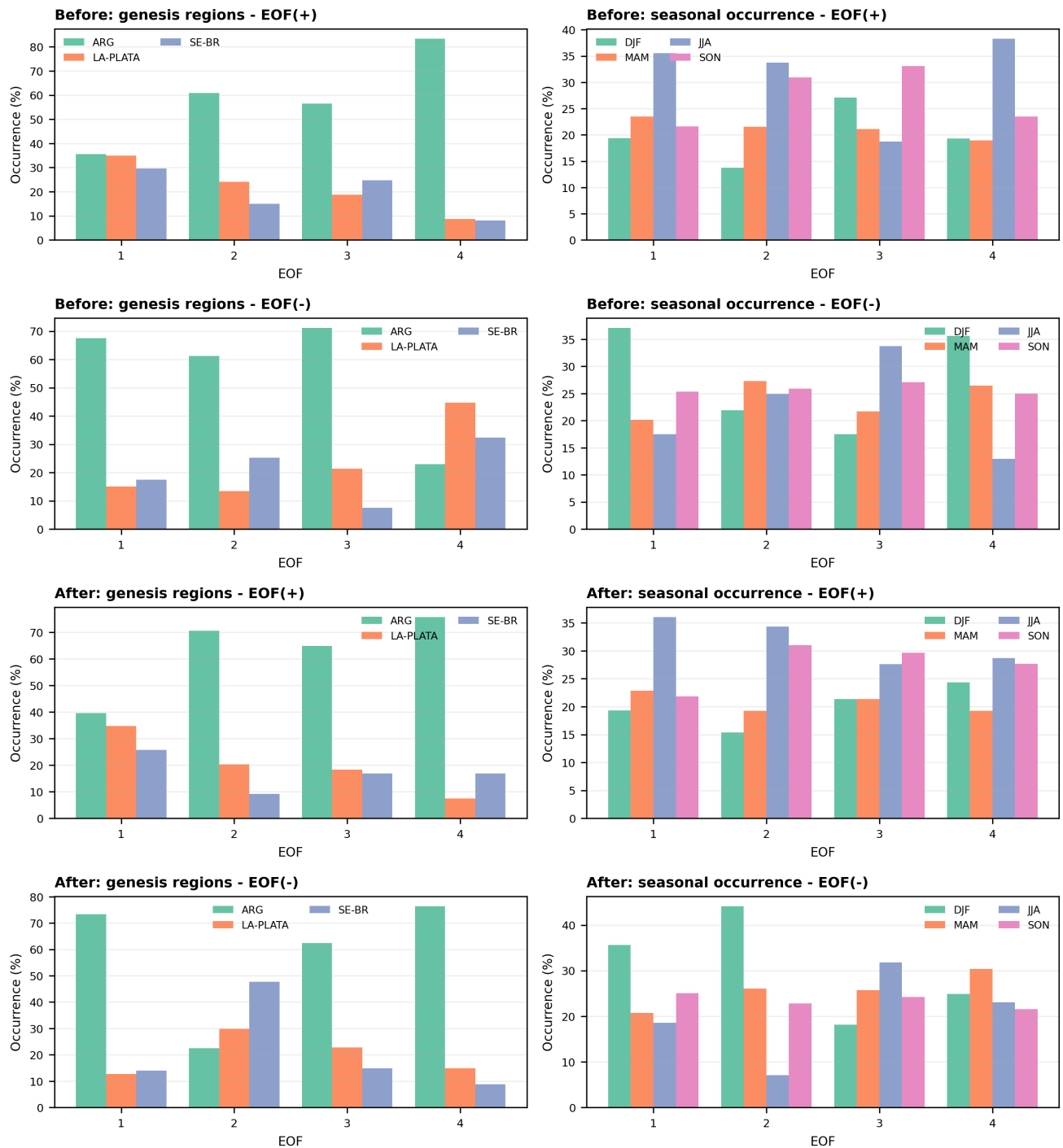
Track density - negative total-lifecycle PC extremes



Negative PC-extreme track densities with before on the upper row and after on the lower row.

Figure 11

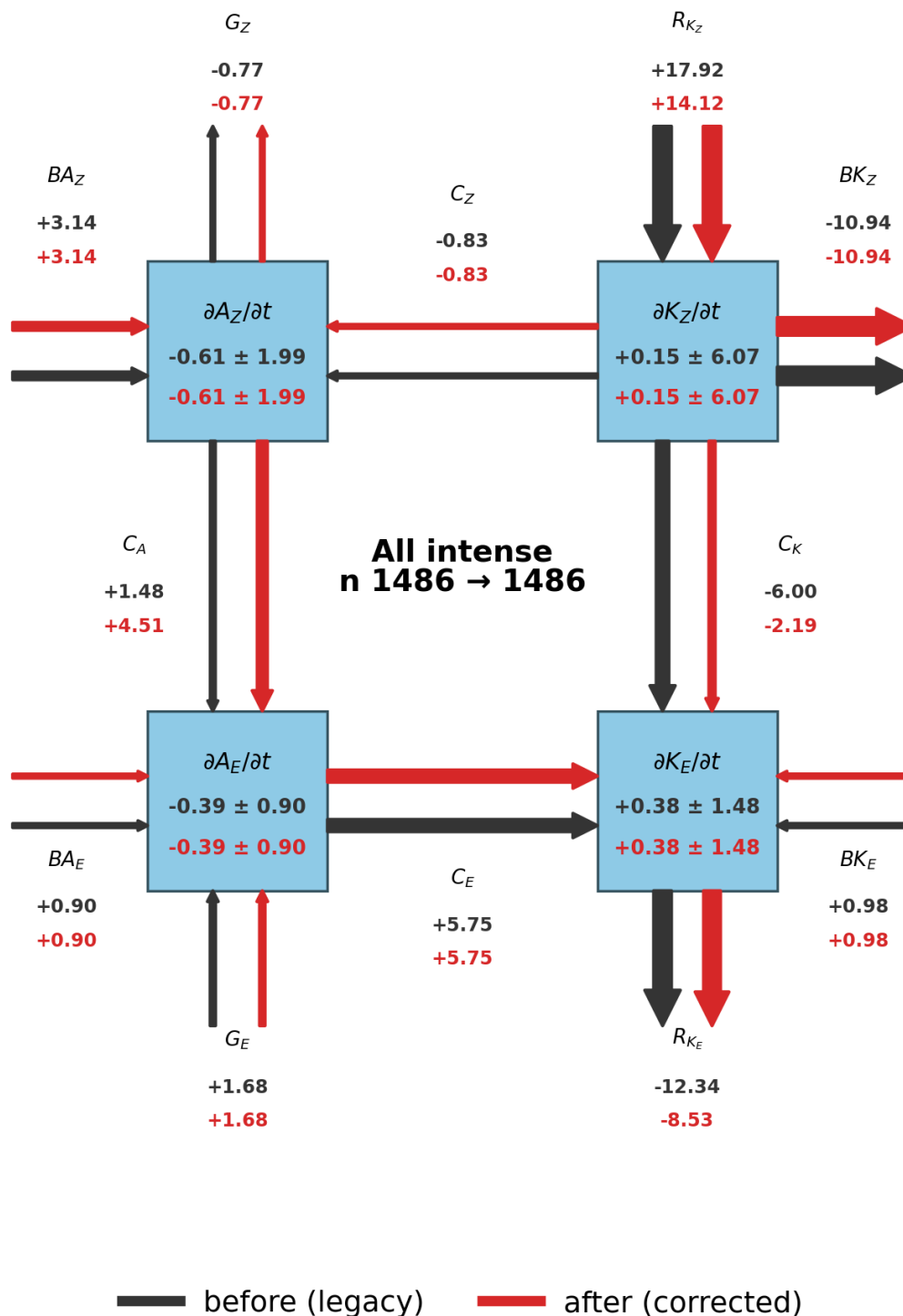
Genesis and seasonal composition of total-lifecycle EOF extremes



Genesis-region and seasonal composition of PC extremes, before above and after below.

Figure 12a

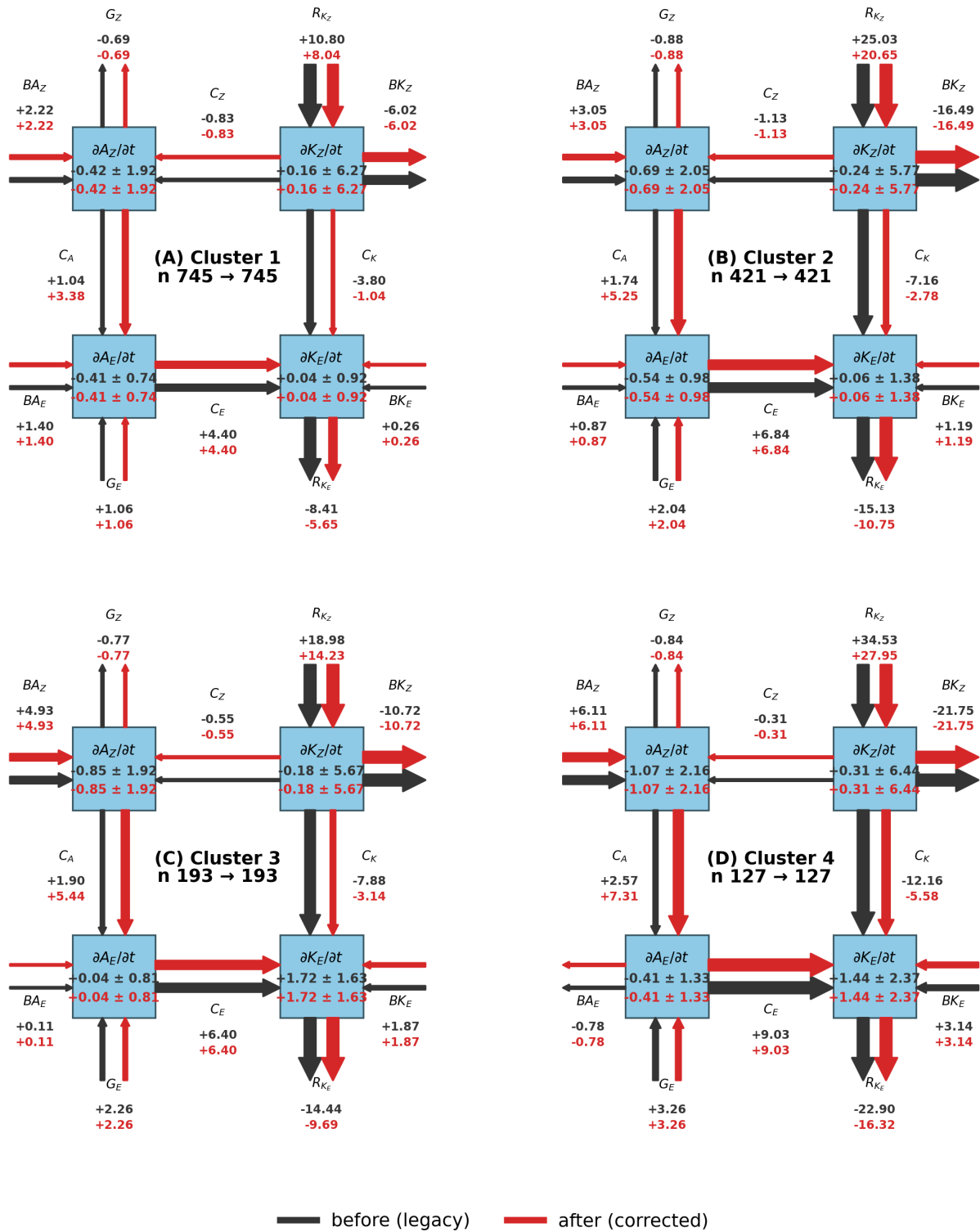
All intense cyclones before and after



All intense cyclones: dark arrows and values are before; red arrows and values are after.

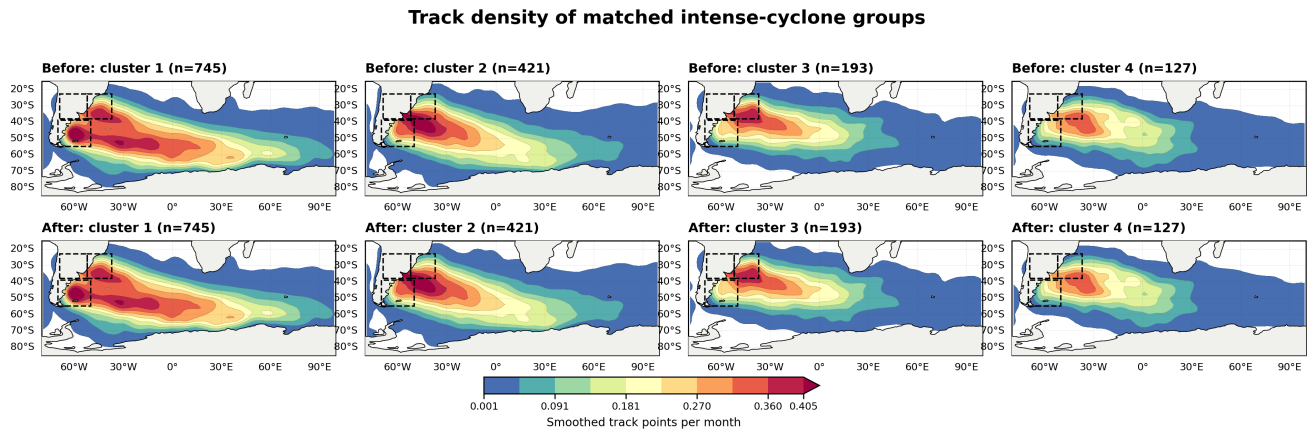
Figure 12b

Matched intense-cyclone clusters before and after



Matched intense-cyclone clusters in a 2x2 layout; dark arrows and values are before and red arrows and values are after.

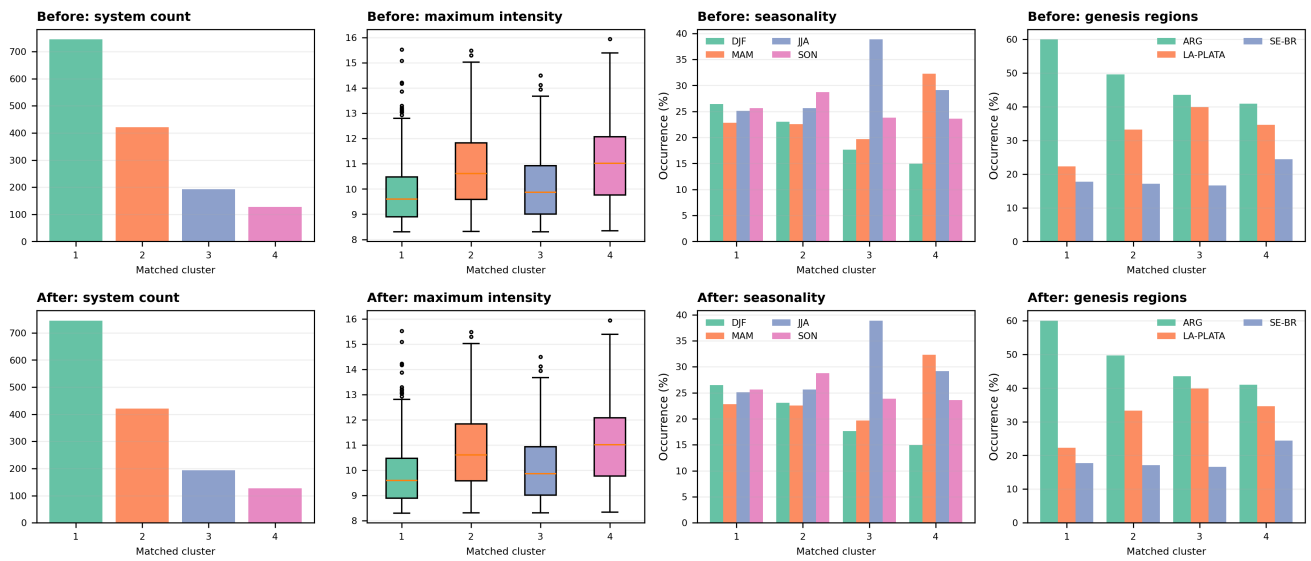
Figure 13



Track densities of intense-cyclone groups, before above and centroid-matched after groups below.

Figure 14

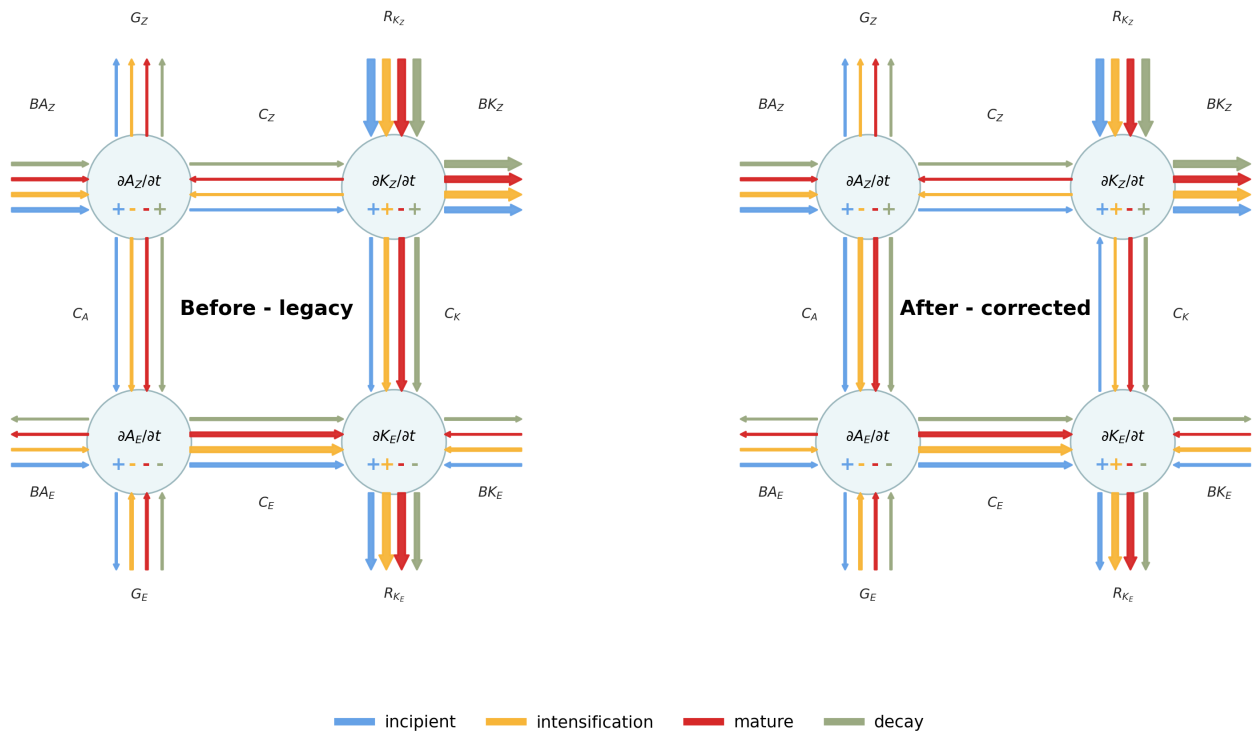
Matched intense-cyclone group characteristics before and after



Intense-group counts, intensity, seasonality and genesis regions, before above and after below.

Figure 15

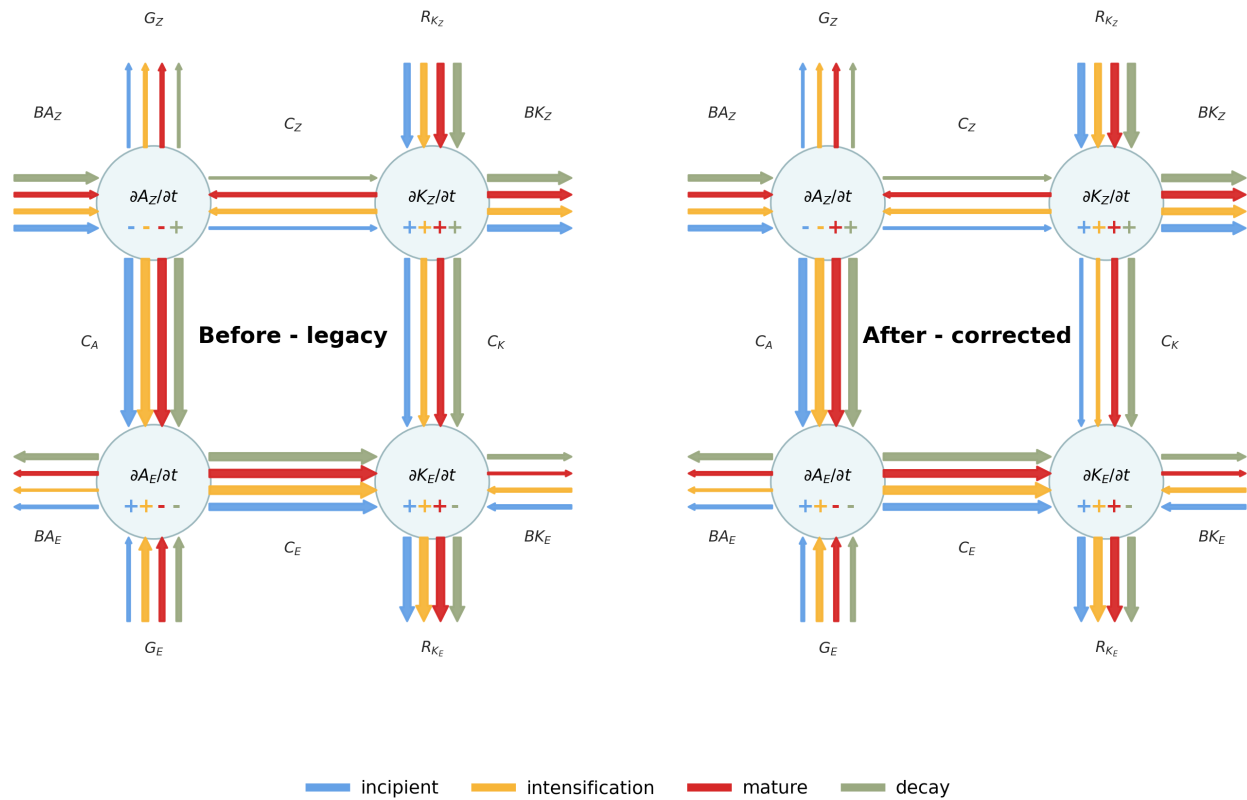
Phase-mean LEC synthesis: before versus after



Lifecycle synthesis shown side by side: before on the left and after on the right.

Figure 16a

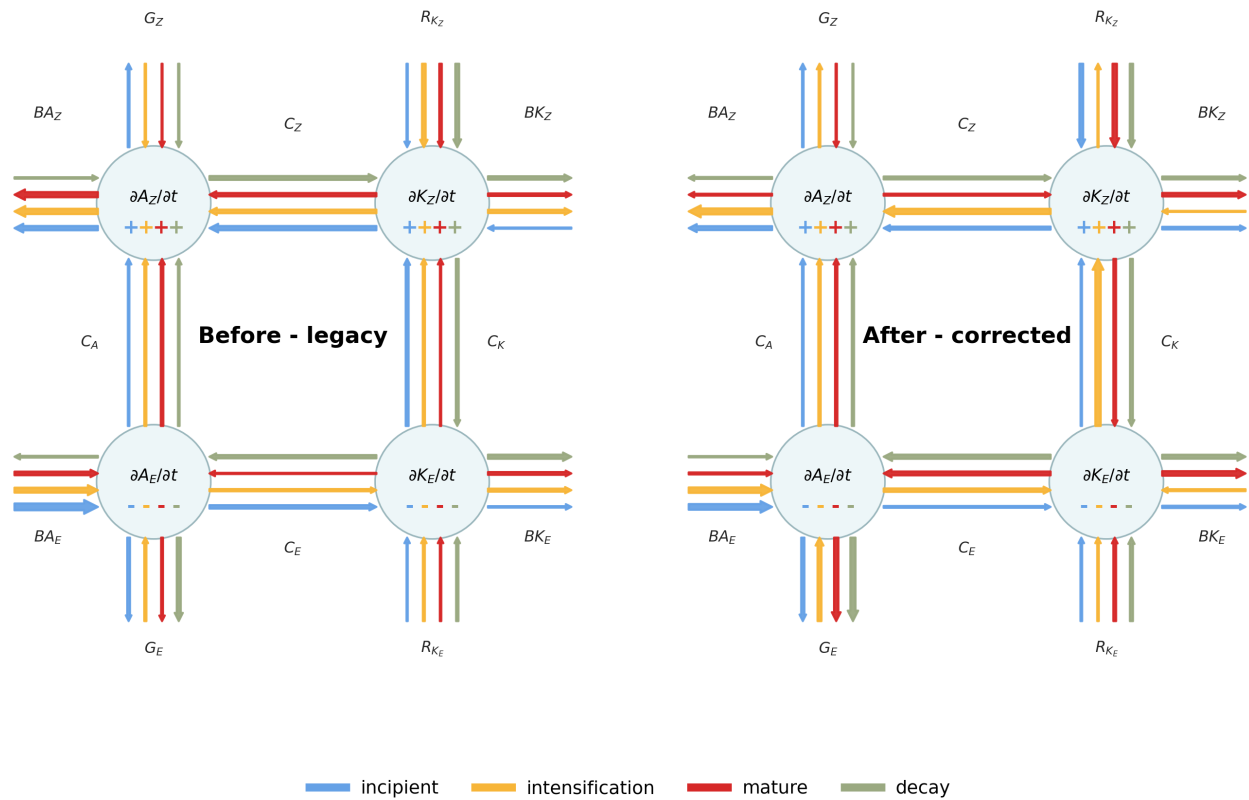
EOF 1 synthesis: before versus matched/sign-aligned after



EOF 1 lifecycle synthesis, before on the left and matched/sign-aligned after on the right.

Figure 16b

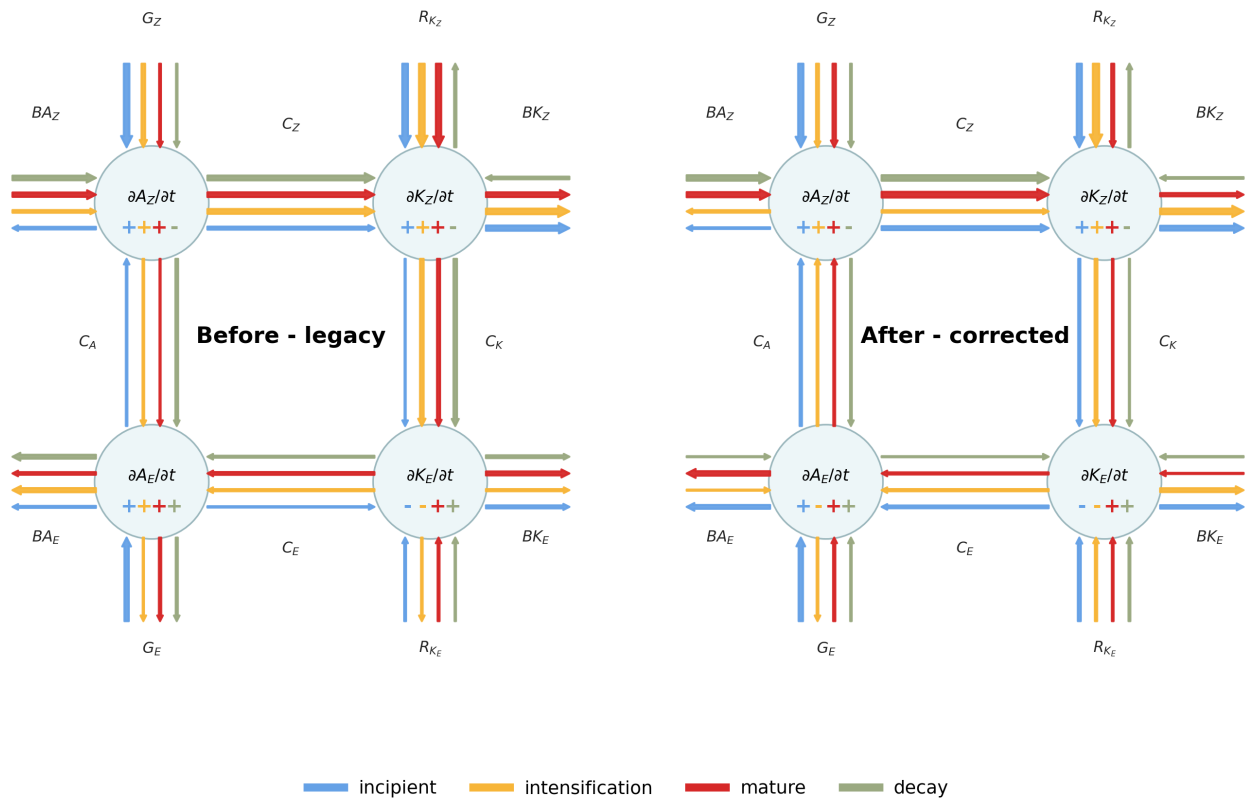
EOF 2 synthesis: before versus matched/sign-aligned after



EOF 2 lifecycle synthesis, before on the left and matched/sign-aligned after on the right.

Figure 16c

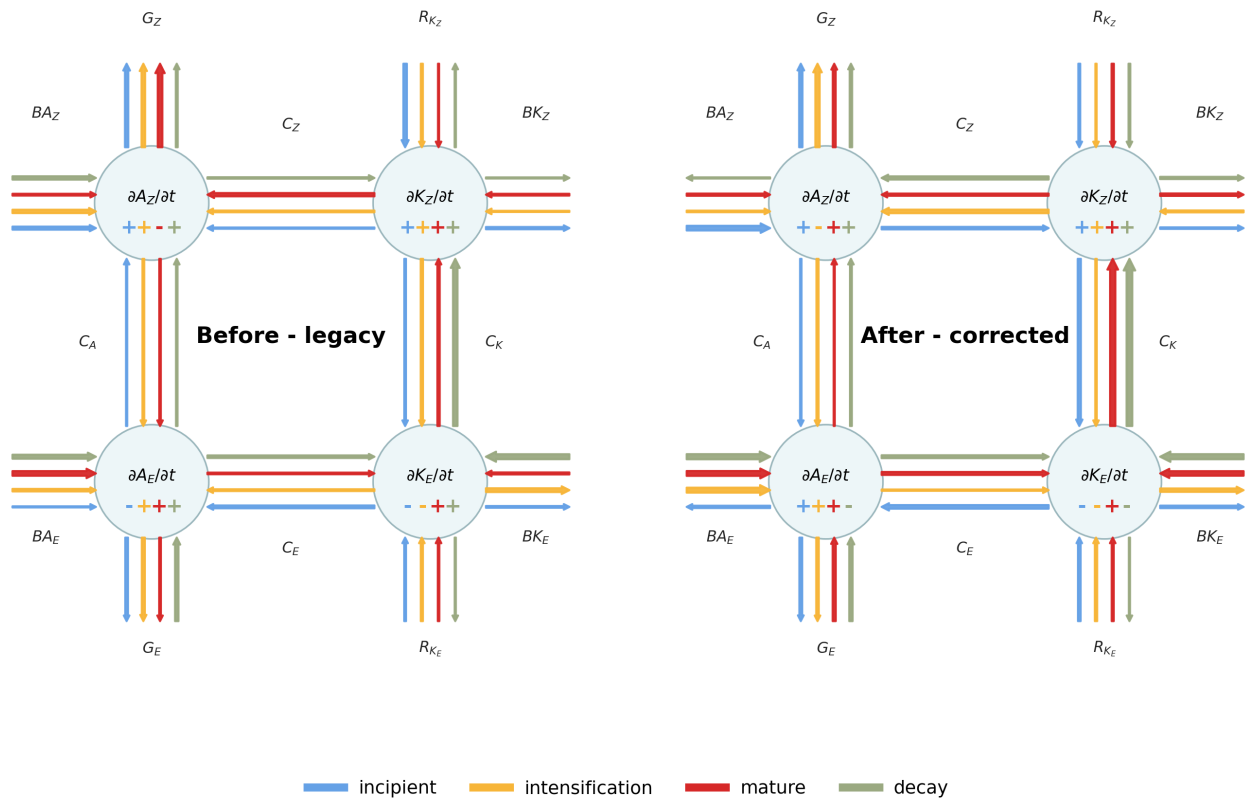
EOF 3 synthesis: before versus matched/sign-aligned after



EOF 3 lifecycle synthesis, before on the left and matched/sign-aligned after on the right.

Figure 16d

EOF 4 synthesis: before versus matched/sign-aligned after



EOF 4 lifecycle synthesis, before on the left and matched/sign-aligned after on the right.

Reproducibility record

Every figure has a PNG and PDF checksum in `results/lec_climatology_corrected/figure_manifest.csv`. The complete input hashes and pinned toolkit commits are recorded in the adjacent `provenance.json`. The workflow is `scripts/article_figures/generate_comparison.py`; this report is built by `scripts/article_figures/build_report.py`.